Notes: Tate & Yang (RFS, 2015)

The Bright Side of Corporate Diversification: Evidence from Internal Labor Markets

Contents

1 Big picture

- **Main question.** What are the benefits of corporate diversification once we look inside the firm at worker allocation, worker mobility, and human-capital development?
- **Core idea.** Diversified firms operate internal labor markets. These markets let firms redeploy workers across business lines when industry opportunities change, and they also create incentives for workers to acquire skills useful across the firm's portfolio of industries.
- **Why this matters.** Much of the diversification literature focuses on internal capital markets and valuation discounts. This paper shifts attention to *internal labor markets*, where diversification may create a bright side even if capital-allocation evidence is mixed.
- **Data advantage.** The paper uses confidential U.S. Census worker-firm-establishment linked data: the Longitudinal Business Database (LBD) and the Longitudinal Employer-Household Dynamics (LEHD) program. This makes it possible to observe workers, wages, firms, establishments, closures, and moves across firms and industries.
- **Firm-level result.** Relative to matched focused firms in the same industries, diversified firms have higher labor productivity. Sales per worker and sales per payroll are higher, especially in industries using more skilled labor.
- **Reallocation result.** After establishment closures, diversified firms retain workers when the firm's remaining industries have stronger growth opportunities, and they move workers out of industries with weaker prospects and toward industries with better prospects.
- **Worker-level result.** Workers from diversified firms who switch industries experience smaller wage losses when they move to industries already spanned by their former diversified firm. They are also more likely to move between the industries that their former firm operates.
- **Mechanism.** Internal labor markets create organization-specific and portfolio-specific human capital. Workers learn skills that are not merely industry-specific, but transferable across the particular bundle of industries operated by the diversified firm.
- **Broad implication.** Higher wages in diversified firms need not be rent dissipation. Higher productivity and stronger outside options can justify higher wages, and a diversification discount can coexist with higher productivity if organization capital is risky.

2 Incremental contribution of the paper

- **Adds labor markets to the diversification debate.** Prior research largely debates whether internal *capital* markets in conglomerates allocate investment efficiently. Tate and Yang add a second internal market: labor. Their contribution is to show that internal labor allocation can be a separate and economically meaningful benefit of diversification.
- **Moves from firm-level outcomes to worker-level mechanisms.** Existing evidence on productivity, wages, and valuation discounts often uses firm- or plant-level data. This paper uses linked employer-employee data to identify the actual movements and wage outcomes of individual workers.
- **Uses establishment closures to separate voluntary from involuntary mobility.** Voluntary job changes can reflect worker preferences, unobserved ability, or private information. The closure sample isolates displaced workers, making cross-firm and cross-industry movements more interpretable as responses to shocks and firm/worker opportunity sets.
- **Provides a new interpretation of higher wages in diversified firms.** Previous work sometimes views higher wages as evidence of agency costs or rent dissipation. This paper shows that workers in diversified firms may be paid more because they acquire transferable skills and have stronger outside options.

- **Reconciles conflicting results on conglomerates.** The paper helps reconcile evidence that diversified firms are more productive with evidence that they may invest less sensitively to industry Q. If labor and capital are partial substitutes, stronger internal labor reallocation can reduce the need for capital reallocation.
- **Connects firm boundaries to human-capital design.** The paper suggests that diversification may be chosen not only because of capital-market frictions, but also because firms can create valuable internal labor markets and broaden workers' human capital.

3 Data and measurement

3.1 LBD: establishments, firms, industries, productivity, and closures

- The Longitudinal Business Database (LBD) covers all U.S. nonfarm establishments with paid employees from 1976 onward.
- It identifies physical establishments, ultimate owners or firms, state and county locations, four-digit SIC industries, employment, payroll, births, and closures.
- The paper uses the LBD to define firm diversification, construct matched focused-firm benchmarks, measure labor productivity, and identify establishment closures.
- A **multi-unit firm** operates at least two establishments.
- A **diversified firm** operates in more than one two-digit SIC industry.
- Establishment closures are central because they generate involuntary displacement events for workers.

Interpretation: The LBD gives the firm-establishment-industrial organization side of the paper. It allows the authors to determine which industries a firm spans and whether a worker's new industry is within or outside the old firm's portfolio.

3.2 LEHD: workers, wages, and worker-firm matches

- The Longitudinal Employer-Household Dynamics (LEHD) data are built from state unemployment insurance records and ES-202 records.
- The LEHD contains individual worker identifiers, firm/unit identifiers, quarterly earnings, age, gender, and race/ethnicity information.
- The data cover 96% of U.S. wage and salary civilian jobs in participating states, but at the time of the paper only 23 states were available through the Census RDC.
- Wages include many forms of compensation reported to unemployment insurance systems, including bonuses and, in some states, deferred compensation contributions.
- The authors compute real quarterly wages and then aggregate to annual real wages using the prior four quarters, because quarterly wage data can contain bonuses and seasonal noise.
- Workers younger than 16 or earning less than $10,000 are excluded.

Interpretation: The LEHD lets the authors observe the same worker before and after displacement, track whether the worker remains with the same firm, changes industries, and experiences wage gains or losses. This is the key innovation that distinguishes the paper from plant-level diversification studies.

3.3 LBD-LEHD matching and closure sample

- For single-unit firms, matching LEHD worker records to LBD establishments is relatively direct because both data systems contain employer identifiers.
- For multi-unit firms, the LEHD reports tax units while the LBD reports physical establishments. The authors use the Census Business Register Bridge at the EIN-state-county-four-digit-SIC level.
- To uniquely match workers to a closing establishment, the LBD establishment must be unique within that bridge partition.
- The authors also require that the LEHD tax unit disappears within a seven-quarter window around the closure observed in the LBD.
- The matched closure sample is smaller than the universe of closing establishments, so the authors are careful about external validity.
- Because workers can reject internal offers and take outside jobs, observed internal reallocation after closures is a lower bound on desired firm reallocation.

Interpretation: The closure-based matched sample is not used simply for convenience. It is a key part of the identification strategy because it helps distinguish involuntary displacement from voluntary mobility.

3.4 Productivity measures and matched focused benchmarks

- Productivity is measured as sales per employee and sales per payroll.
- For diversified firms, each segment is matched to focused firms in the same two-digit SIC industry, year, firm-size bin, and age bin.
- The focused benchmarks are used to predict what the diversified firm's sales would be if each segment had focused-firm productivity.
- The paper then compares actual diversified-firm sales with the predicted sales from matched focused firms.
- In manufacturing subsamples, the authors use shipments, capital, and materials variables in a more traditional plant productivity specification.

Interpretation: Matching by industry, size, and age is designed to separate diversification from scale, lifecycle, and industry-composition effects. The productivity test asks whether diversified firms produce more output than comparable focused firms using similar labor inputs.

3.5 Industry opportunities and high-skill measures

- Industry opportunities are proxied by the median Tobin's Q of single-segment public firms in the same two-digit SIC industry.
- Industry Q is computed using Compustat as market value of assets divided by book value of assets.
- Expected changes in opportunities are measured with changes in industry Q following the establishment closure window.
- High-skill industries are defined using two-digit SOC codes: an industry is high-skill if its share of workers in highly skilled occupations exceeds the sample median.
- The high-skill classification matters because broad, transferable human-capital investments should be more valuable when workers have complex or scarce skills.

Interpretation: The central empirical prediction is not only that diversified firms move workers, but that they move workers in response to changing opportunity sets. Industry Q provides the external measure of those opportunity sets.

3.6 Spanned industries and worker-level mobility

- A worker's post-closure new industry is called a **spanned industry** if the worker's old diversified firm operated in that industry, excluding the worker's own old industry.
- The paper compares wage changes for workers moving to spanned industries versus unspanned industries.
- This test is designed to detect whether diversified-firm workers acquired skills that are useful in other industries within the firm's portfolio.
- The paper also tests whether workers from diversified firms are more likely to move across industry pairs that their old firm spans.

Interpretation: The spanned-industry concept is the key worker-level mechanism variable. If a worker leaves the firm but moves into an industry the former firm operated, better outcomes suggest that internal labor-market exposure created externally valuable skills.

4 Empirical specifications and models

4.1 Productivity benchmark for diversified firms

For each segment of a diversified firm, the authors form a benchmark using focused firms in the same two-digit SIC industry, year, age bin, and size bin. Let actual sales be $Sales_{jt}$ and predicted sales from matched focused firms be $\widehat{Sales}_{jt}$. The main productivity outcome is excess sales:

$$ExcessSales_{jt} = \ln(Sales_{jt}) - \ln(\widehat{Sales}_{jt}).$$

The baseline regression is:

$$ExcessSales_{jt} = \alpha + \beta Diversified_{jt} + \gamma_1 \ln(Age_{jt}) + \gamma_2 \ln(Emp_{jt}) + \eta_{industry \times year} + \varepsilon_{jt}.$$

Interpretation: A positive β means diversified firms generate more sales than comparable focused firms with similar industry, size, and age characteristics. Because the dependent variable is log excess sales, coefficients can be interpreted approximately as percentage differences.

4.2 Heckman selection model for firm-initiated redeployment

For workers displaced by closures in diversified firms, the authors model two decisions:

1. the firm retains the worker inside the same firm;
2. conditional on being retained, the firm reallocates the worker to a different two-digit SIC industry.

The first stage is a probit for same-firm retention. The second stage is a linear probability model for industry change including the inverse Mills ratio:

$$Pr(SameFirm_{it} = 1) = \Phi(Z'_{it}\pi),$$

$$IndustryChange_{it} = X'_{it}\beta + \rho\lambda_{it} + u_{it} \quad \text{conditional on } SameFirm_{it} = 1.$$

Interpretation: The two-stage design recognizes that industry changes inside a diversified firm are observed only for workers retained by the firm. The selection correction helps distinguish the decision to keep the worker from the decision to redeploy the worker across industries.

4.3 All-worker industry-switching regression

The paper next studies all displaced workers who find jobs by quarter $t + 3$:

$$IndustryChange_{it} = \alpha + \beta_1 Diversified_i + \beta_2 ChgQ_i + \beta_3 (Diversified_i \times ChgQ_i) + X_i' \Gamma + heta_s + heta_j + heta_t + \varepsilon_i.$$

Interpretation: The coefficient on $Diversified \times ChgQ$ asks whether workers from diversified firms are more likely to leave an industry when that industry has poor future prospects. A negative coefficient means high future Q in the old industry lowers industry switching among diversified-firm workers.

4.4 New-versus-old industry opportunity regression

For displaced workers who switch industries, the authors compute:

$$\Delta Q_{new-old} = \Delta \ln Q_{new} - \Delta \ln Q_{old}.$$

They estimate:

$$\Delta Q_{new-old,i} = \alpha + \beta SameFirm_i + \gamma Diversified_i + X_i' \Gamma + heta_{old\ industry} + \varepsilon_i.$$

Interpretation: The coefficient on $SameFirm$ captures internal moves in diversified firms. A positive coefficient means internal switchers move into industries with better expected growth than comparable external market switchers.

4.5 Worker wage-change regression for spanned industries

The main wage outcome is the change in annual real log wages between quarter $t - 2$ and quarter $t + 4$ around establishment closure:

$$\Delta \ln Wage_{i,t+4,t-2} = \ln(Wage_{i,t+4}) - \ln(Wage_{i,t-2}).$$

The key specification is:

$$\Delta \ln Wage_i = \alpha + \beta_1 DifferentIndustry_i + \beta_2 SameFirm_i + \beta_3 (SameFirm_i \times DifferentIndustry_i) + \beta_4 (DifferentIndustry_i$$

Interpretation: The coefficient on $DifferentIndustry \times SpannedIndustry$ measures whether workers from diversified firms suffer smaller wage losses when they move to industries their former firm also operated. The inclusion of SIC-pair fixed effects in the strongest specification compares workers moving between the same old-new industry pairs.

4.6 Heterogeneity: skill, tenure, and prior experience

The paper extends the wage-change regression by interacting the spanned-industry term with high-skill indicators, low-tenure indicators, and prior industry experience:

$$\Delta \ln Wage_i = \cdots + \beta_4 (DifferentIndustry_i \times SpannedIndustry_i) + \beta_5 (HighSkill_i \times DifferentIndustry_i \times SpannedIndustry_i)$$

Interpretation: These tests help distinguish three mechanisms: workers may have pre-existing skills, firm-created skills, or industry-pair-specific experience. The evidence points toward skills developed inside diversified firms, because effects are stronger for high-skill workers and for workers with longer tenure, and are not explained by prior experience in the destination industry.

5 Identification strategy

5.1 Identification problem

The core empirical challenge is that diversified and focused firms differ in many ways: size, worker composition, sector mix, age, labor demand shocks, and worker self-selection. Voluntary job changes are also endogenous. A worker who chooses to move may be more able, may have better private information, or may be responding to personal preferences rather than firm reallocation.

5.2 Strategy 1: matched focused-firm productivity benchmarks

For the productivity analysis, the authors compare diversified-firm segments to focused firms in the same industry-year-size-age cells. They then add controls for age and employment and industry-year fixed effects.

Interpretation: This strategy reduces mechanical differences driven by industry composition and size. It does not fully solve selection into diversification, but it asks whether diversified firms are more productive than an observably close focused-firm benchmark.

5.3 Strategy 2: establishment closures as displacement events

The worker mobility tests focus on workers displaced by establishment closures. This follows the logic of Gibbons and Katz: closures generate involuntary job loss and reduce the confounding from voluntary worker mobility.

Interpretation: If the worker leaves because the establishment shuts down, the subsequent job change is less likely to reflect the worker's private choice to move at exactly that time. This makes wage changes and industry transitions more comparable across workers.

5.4 Strategy 3: internal versus external reallocation after closures

Among displaced workers, the paper distinguishes:

- workers retained by the same diversified firm;
- workers leaving the firm and moving in the external market;
- workers changing industries;
- workers moving to spanned versus unspanned industries.

The internal moves provide a direct test of whether diversified firms redeploy workers toward stronger industries. External moves to spanned industries test whether the worker's human capital is useful outside the firm.

5.5 Strategy 4: fixed effects and narrow comparisons

The wage regressions include combinations of state, industry, year, establishment, and SIC-pair fixed effects.

- State fixed effects absorb geographic labor-market differences.
- Industry fixed effects absorb differences in source industries.
- Year fixed effects absorb aggregate labor-market conditions.
- Establishment fixed effects compare workers displaced from the same closing establishment.
- SIC-pair fixed effects compare workers making the same old-industry to new-industry transition.

Interpretation: The strongest wage specifications ask whether, among workers moving between the same pair of industries, those whose former diversified firm spanned the destination industry experience smaller wage losses.

5.6 Strategy 5: mechanism tests against alternative explanations

The paper tests whether the spanned-industry wage result is concentrated among high-skill workers, rises with tenure in the diversified firm, and survives controls for prior experience in the destination industry. These are mechanism tests.

Interpretation: If the results were only sorting, workers with prior experience in the destination industry should explain the effect. If the results are about firm-created portfolio skills, the effect should grow with tenure and be stronger for high-skill workers. The evidence supports the latter interpretation.

5.7 Remaining threats and limits

- LEHD coverage is limited to 23 states in the analysis period.
- The LBD-LEHD matched closure sample is smaller and may underrepresent complex multi-unit firms.
- Establishment closures reduce voluntary mobility bias but may still reflect firm-level distress or selection.
- Industry Q is an imperfect proxy for opportunities, especially for private firms and nonpublic industries.
- Diversification itself is not randomly assigned, so cross-sectional productivity differences should be interpreted cautiously.
- The paper is strongest on mechanisms around worker mobility and wage changes after closures; it is more suggestive on valuation and broader diversification decisions.

6 Tables and figures

6.1 Figure 1: States covered in the Longitudinal Employer-Household Dynamics (LEHD) data

Read:

- The map displays the 23 states for which worker-firm matched LEHD data are available in the research period.
- Covered states include many regions, but important states such as New York are missing.
- Missing states matter because workers who move to uncovered states can disappear from the data.

Interpretation: The map is a data-validity figure. It shows that the analysis is based on a large but incomplete subset of the U.S. labor market. The authors argue that migration to new covered states is low and that most of the analysis focuses on wage changes, but coverage remains an external-validity issue.

Economic message: The paper's credibility depends on transparent use of administrative data and careful acknowledgement of what the LEHD does and does not cover.

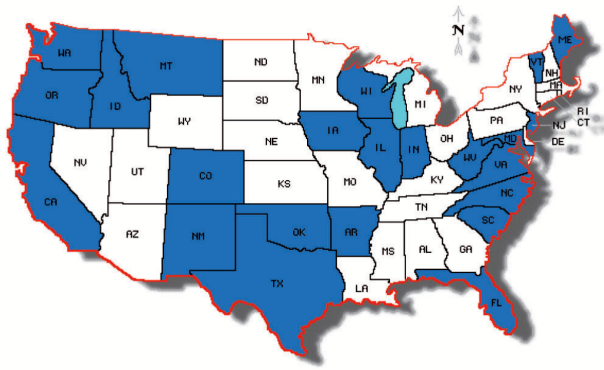

Figure 1
States covered in the Longitudinal Employer-Household Dynamics (LEHD) data
The map shows states for which worker-firm matched panel data are available through the U.S. Census Bureau's LEHD program. No data are available for unshaded states.

6.2 Table 1: Summary statistics - plant level

Read:

- The random LBD sample has 655,929 establishments; the LBD closure sample has 143,370 establishments; the matched LBD-LEHD closure sample has 12,439 establishments.
- The matched closure sample contains smaller establishments and smaller firms than the broader LBD closure sample.
- In the full random LBD sample, 58% of establishments are part of multi-unit firms and 42% are part of diversified firms.
- Among multi-unit firms, diversified firms account for 71% of random establishments and 69% of matched closing establishments.

Interpretation: The matched worker-establishment sample is clearly selective, because it requires a bridge from LEHD tax units to LBD establishments. The key question is not whether the matched sample is identical to the full universe, but whether the selection is plausibly unrelated to the wage-change and internal-reallocation mechanisms being studied.

Economic message: Even in the matched sample, many multi-unit closing establishments belong to diversified firms, so the data retain the variation needed to study internal labor markets.

6.3 Table 2: Summary statistics - worker level

Read:

- In the random LEHD sample, the average annual wage is $34,999; diversified-firm workers earn $37,121 on average.
- The random worker sample has 251,440 observations; the matched closure worker sample has 461,449 observations.
- Workers in the matched closure sample earn less on average than workers in the random sample, consistent with closures occurring at smaller or weaker establishments.

Table 1
Summary statistics: Plant level

	Panel A: All firms			Panel B: Multi-unit firms only		
	Random establishments in the LBD (N=655,929)	Closing establishments in the LBD (N=143,370)	Closing establishments in the LBD matched with the LEHD (N=12,439)	Random establishments in the LBD (N=383,238)	Closing establishments in the LBD (N=70,811)	Closing establishments in the LBD matched with the LEHD (N=1,850)
Plant employees	194	188	134	202	187	142
	(514)	(647)	(292)	(473)	(565)	(224)
Firm employees	25,765	22,084	4,780	43,968	44,521	31,379
	(83,464)	(57,124)	(26,992)	(105,480)	(74,912)	(63,789)
Annual payroll ($000's)	$6,830	$5,299	$2,333	$7,590	$6,676	$3,703
	($383,230)	($66,606)	($6,709)	($178,102)	($92,809)	($9,611)
% of Multi-unit firms	0.58	0.49	0.15			
% of diversified firms	0.42	0.39	0.10	0.71	0.79	0.69
Industry distribution						
SIC = 1	0.05	0.04	0.09	0.02	0.02	N/A
SIC = 2	0.08	0.08	0.08	0.09	0.08	N/A
SIC = 3	0.10	0.08	0.07	0.10	0.09	N/A
SIC = 4	0.06	0.07	0.05	0.08	0.08	N/A
SIC = 5	0.29	0.27	0.28	0.36	0.30	N/A
SIC = 6	0.06	0.09	0.04	0.07	0.10	N/A
SIC = 7	0.13	0.19	0.24	0.13	0.18	N/A
SIC = 8	0.21	0.16	0.13	0.15	0.13	N/A
Geographic distribution						
LEHD State	0.55	0.57	.	0.55	0.57	.
Region = NE	0.22	0.22	0.08	0.21	0.22	0.09
Region = MW	0.25	0.21	0.16	0.25	0.22	0.18
Region = S	0.23	0.24	0.23	0.24	0.24	0.26
Region = SW	0.12	0.13	0.19	0.12	0.12	0.19
Region = W	0.14	0.16	0.29	0.14	0.15	0.22
Region = RM	0.04	0.03	0.05	0.04	0.03	0.06
Yearly distribution						
Year = 1994	0.10	0.08	0.08	0.10	0.07	0.05
Year = 1995	0.11	0.08	0.10	0.10	0.08	0.07
Year = 1996	0.11	0.11	0.12	0.11	0.11	0.13
Year = 1997	0.11	0.10	0.09	0.11	0.10	0.07
Year = 1998	0.11	0.11	0.13	0.12	0.11	0.12
Year = 1999	0.12	0.12	0.12	0.12	0.14	0.10
Year = 2000	0.12	0.12	0.14	0.12	0.13	0.22
Year = 2001	0.12	0.21	0.14	0.12	0.17	0.17

Panel A reports summary statistics of a random sample of nonclosing establishments from the LBD, all closing establishments in the LBD, and the subsample of closing establishments from the LBD that we match with worker-level data from the LEHD program. The random sample of LBD establishments includes 20% of LBD establishments during the sample period 1993-2001. Panel B reports the corresponding statistics for the subsamples of establishments from multi-unit firms. We define multi-unit firms as firms that operate at least two distinct establishments. Standard deviations are reported in parantheses for continuous variables. N/A indicates industries that have a limited number of firms. Because of potential disclosure risk, we cannot report the industry distribution for these subsamples.

Table 2
Summary statistics: Worker level

Panel A: Random workers from the LEHD data

	Full sample (N = 251,440)	Single-unit firms (N = 63,173)	Multi-unit focused firms (N = 34,042)	Multi-unit diversified firms (N = 154,225)
Annual wage	$34,999	$30,613***	$33,527***	$37,121
	(92,402)	(64,364)	(93,173)	(101,461)
Age	41.33	42.59***	40.06***	41.09
	(11.10)	(11.28)	(11.30)	(10.94)
Tenure (in yrs)	3.36	3.49***	3.17***	3.34
	(2.61)	(2.68)	(2.52)	(2.59)
Education (in yrs)	13.79	13.89***	13.73**	13.76
	(2.60)	(2.60)	(2.63)	(2.59)
% of Female	0.46	0.51***	0.49***	0.43
Race = Black	0.10	0.10**	0.10	0.10
Race = Asian	0.04	0.03***	0.04***	0.04
Race = Hispanic	0.09	0.10***	0.09***	0.08
Race = Other	0.05	0.05	0.06*	0.05
% of Foreigner	0.14	0.14*	0.15***	0.14

Panel B: Workers from the LEHD data matched to closing establishments from the LBD

	Full sample (N = 461,449)	Single-unit firms (N = 395,338)	Multi-unit focused firms (N = 15,947)	Multi-unit diversified firms (N = 50,137)
Annual wage	$29,933^{a}	$29,751^{***,a}	$28,642^{***,a}	$31,781^{a}
	(54,517)	(56,278)	(33,666)	(44,897)
Age	39.68^{a}	39.53^{***,a}	39.59^{***,a}	40.89^{a}
	(11.43)	(11.47)	(11.53)	(10.99)
Tenure (in yrs)	2.57^{a}	2.52^{***,a}	2.69^{***,a}	2.96^{a}
	(2.20)	(2.18)	(2.51)	(2.17)
Education (in yrs)	13.66^{a}	13.64^{***,a}	13.64^{***,a}	13.82^{a}
	(2.66)	(2.67)	(2.60)	(2.60)
% of Female	0.41^{a}	0.41^{*,a}	0.42^{a}	0.41^{a}
Race = Black	0.10	0.10***	0.13^{***,a}	0.11^{a}
Race = Asian	0.04^{a}	0.04^{***,a}	0.05^{***,a}	0.04
Race = Hispanic	0.12^{a}	0.13^{***,a}	0.10^{***,a}	0.09^{a}
Race = Other	0.06^{a}	0.06^{***,a}	0.05*	0.05^{a}
% of Foreigner	0.19^{a}	0.19^{***,a}	0.18^{***,a}	0.15^{a}

Panel A reports summary statistics for a random sample of workers from the LEHD data. The random sample of the LEHD data includes 0.4% of workers during the sample period 1993-2001. Panel B reports summary statistics for workers matched to closing establishments in the LBD. We report statistics for the overall sample and for the subsmples of worker from single-unit firms, multi-unit focused firms, and multi-unit diversified firms. We define multi-unit firms as firms which operate at least two distinct establishments and diversified firms as firms which operate in more than one two-digit SIC code. Standard deviations are reported in parantheses for continuous variables. ***, **, or * indicate a significance of the difference in means between workers in the given column (single-unit or multi-unit focused firms) and workers in multi-unit diversified firms at the 1%, 5%, and 10% levels, respectively. a, b, or c in Panel B indicate significance of the cross-sample differences in means within groups between workers in the closing sample and workers from the random sample at the 1%, 5%, and 10% levels, respectively.

- Diversified-firm workers in the closure sample still earn more than workers in single-unit or focused multi-unit firms.

Interpretation: Higher wages in diversified firms are not automatically evidence of inefficiency. They could reflect higher skills, more complex internal labor markets, better outside options, or compensation for workers who invest in broader human capital.

Economic message: Table 2 motivates why worker-level controls and closure-based identification are necessary: worker composition differs across organizational forms.

6.4 Figure 2: Firm size within match cohorts

Read:

- Panel A shows the distribution of t-statistics for employment differences between focused and diversified firms within match cells.
- Most mass is near zero, indicating that the matching procedure generally balances firm size.
- Panel B shows that large imbalances are concentrated mostly among very small firms and very large firms.

Interpretation: This figure supports the productivity matching design. It documents whether diversified and focused firms in the matched cells are similar in employment. Remaining imbalance motivates continuous controls for size and age in the regression specifications.

Economic message: Matching quality matters because productivity differences could otherwise be driven by scale economies rather than diversification.

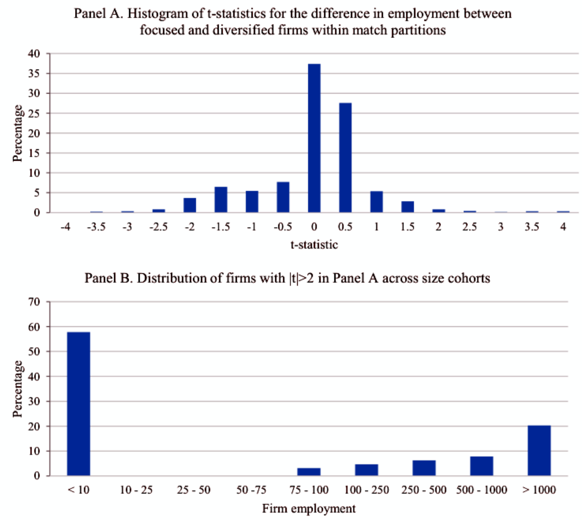

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Table 3
Productivity and cash flows in diversified firms

	Panel A. Sales/employment		Panel B. Sales/payroll		Panel C. Manufacturing firms	
	(1)	(2)	(3)	(4)	(5)	(6)
Ln(firm age)	−0.079***	−0.079***	−0.032***	−0.032***	−0.025***	−0.024*
	(0.004)	(0.004)	(0.002)	(0.002)	(0.003)	(0.003)
Ln(firm employment)	−0.196***	−0.196***	−0.083***	−0.083***	−0.688***	−0.688*
	(0.004)	(0.004)	(0.002)	(0.002)	(0.004)	(0.004)
Ln(capital stock)					0.107***	0.107*
					(0.004)	(0.004)
Ln(material and energy costs)					0.534***	0.534*
					(0.005)	(0.005)
Diversified	***0.265****	***0.174****	***0.163****	***0.148****	***0.076****	***0.066****
	(0.016)	(0.018)	(0.008)	(0.013)	(0.004)	(0.004)
% emp. in high-skill ind.		0.145***		−0.024		0.050*
		(0.044)		(0.040)		(0.009)
Diversified × % emp. in high-skill ind		***0.275****		***0.044****		***0.025****
		(0.022)		(0.023)		(0.008)
Industry × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.135	0.135	0.035	0.035	0.771	0.771
N	3,601,677	3,601,677	3,601,677	3,601,677	418,848	418,848

The sample in Panels A and B consists of all firm-years for which we can match each individual segment (defined by 2-digit SIC codes) to a focused firm benchmark, excluding firm

6.5 Table 3: Productivity and cash flows in diversified firms

Read:

- In sales per employment specifications, the diversified-firm coefficient is 0.265 without skill interaction and 0.174 with skill interaction.
- In sales per payroll specifications, the diversified coefficient is positive, around 0.163 to 0.148.
- The interaction between diversification and high-skill industry employment is positive and significant.
- In manufacturing firms, diversified establishments also show higher productivity relative to focused benchmarks.

Interpretation: The table shows that diversified firms are not merely paying more labor costs. They have higher output per worker and output per payroll, and the effect is stronger where skilled labor is important. This supports the idea that internal labor markets and human capital matter for productivity.

Economic message: Diversification can create productive synergies through human capital, not only through capital-market allocation.

6.6 Table 4: Labor redeployment - diversified firms

Read:

- Column 1 models whether displaced workers remain with the same diversified firm after establishment closure.
- Column 2 models whether retained workers change industries inside the firm.
- Higher firm Q is positively related to retention, while stronger prospects in the worker's old industry reduce industry changes.

- The selection term is significant, confirming that retention and redeployment are not independent decisions.

Interpretation: Diversified firms retain workers when they have stronger remaining opportunities and redeploy workers away from weaker old industries. The results are consistent with internal labor markets functioning as reallocation mechanisms.

Economic message: Diversified firms use internal labor markets similarly to how internal capital-market theories predict firms reallocate physical capital.

Table 4
Labor redeployment: Diversified firms

	Heckman selection model	
Dependent variable	Same firm (1)	Industry change (2)
Ln(wage)	0.289***	0.095***
	(0.020)	(0.010)
Ln(age)	0.011	0.003
	(0.035)	(0.014)
Race = Black	0.044	−0.027**
	(0.029)	(0.011)
Race = Asian	0.216***	0.043**
	(0.045)	(0.018)
Race = Hispanic	0.128***	0.016
	(0.033)	(0.013)
Race = Other minorities	0.064	0.003
	(0.042)	(0.017)
Female	0.140***	0.037***
	(0.021)	(0.009)
Ln(tenure)	−0.018	0.000
	(0.013)	(0.005)
Manager	−0.184**	0.007
	(0.088)	(0.037)
N_Estabs	0.009***	0.012***
	(0.002)	(0.001)
Ln(EstabEmp)	0.231***	−0.007
	(0.014)	(0.009)
Ln(FirmEmp)	−0.047***	−0.002
	(0.008)	(0.004)
Native to state	−0.002	−0.026***
	(0.020)	(0.008)
Ln(# of firms in CT & SIC2)	0.854***	0.263***
	(0.030)	(0.023)
Ln(# of firms in CT)	−0.617***	−0.125***
	(0.029)	(0.020)
Chg_Q	−0.452***	−0.387***
	(0.071)	(0.037)
Firm_Q	0.680***	
	(0.033)	
State fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Lambda		0.389***
N	36,244	8,025

The sample contains one observation for each worker displaced from closing establishments of diversified firms. Column 1 presents coefficient estimates from a probit regression. Column 2 is the second stage of a Heckman selection model, with the regression in Column 1 serving as the first stage. The second stage is estimated as a linear probability model. The dependent variable is indicated in the column header. Same firm equals one if the worker remains in the same firm after the establishment closure and zero otherwise. Industry change equals one if the new job in quarter t + 4 is in a different 2-digit SIC from the lost job. Ln(wage) is the natural log of the annualized wage. Ln(age) is the natural log of the worker's age. Race = "*x*" is an indicator variable that equals one for workers of race "*x*" and zero otherwise. Female is an indicator variable that equals one for female workers and zero otherwise. Ln(tenure) is the natural log of the number of quarters that a worker has spent in the SEIN. Manager is an indicator variable equal to one for the highest paid employee in the SEIN and zero otherwise. N_Estabs is the number of establishments owned by the firm, divided by 100. Ln(EstabEmp) is the natural log of establishment employment. Ln(FirmEmp) is the natural log of aggregate firm employment. Native to State

Table 5
Labor redeployment: All workers

Dependent variable: Industry change	(1)	(2)
Ln(wage)	−0.045***	−0.044***
	(0.007)	(0.006)
Ln(age)	−0.093***	−0.093***
	(0.007)	(0.007)
Race = Black	0.007	0.008
	(0.007)	(0.007)
Race = Asian	−0.009	−0.008
	(0.011)	(0.011)
Race = Hispanic	−0.019***	−0.018***
	(0.007)	(0.007)
Race = Other Minorities	−0.013***	−0.013***
	(0.005)	(0.005)
Female	0.007	0.008*
	(0.004)	(0.004)
Ln(tenure)	−0.047***	−0.047***
	(0.004)	(0.004)
Manager	0.037***	0.037***
	(0.009)	(0.009)
Multi-unit	0.060**	0.076**
	(0.029)	(0.029)
N_plants	0.003	0.002
	(0.002)	(0.002)
Ln(EstabEmp)	−0.045***	−0.047***
	(0.009)	(0.009)
Ln(FirmEmp)	0.008	0.010
	(0.008)	(0.008)
Native to state	0.002	0.001
	(0.003)	(0.003)
Ln(# of firms in CT & SIC2)	−0.062***	−0.060***
	(0.011)	(0.011)
Ln(# of firms in CT)	0.062***	0.060***
	(0.010)	(0.010)
Diversified	0.007	0.028
	(0.037)	(0.038)
Chg_Q	−0.012	−0.014
	(0.021)	(0.021)
Same firm		−0.198***
		(0.048)
Chg_Q × Diversified	−0.153*	−0.184**
	(0.083)	(0.082)
Divers × Same firm × Chg_Q		0.040
		(0.225)
State fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
R^2	0.134	0.137
N	343,206	343,206

The sample contains one observation for each worker displaced from a closing establishment in our matched LBD-LEHD data. The estimates are from a linear probability model. The dependent variable, Industry change, equals one if the new job in quarter t + 4 is in a different 2-digit SIC from the lost job. Ln(wage) is the natural log of the annualized wage. Ln(age) is the natural log of the worker's age. Race = "*x*" is an indicator variable that equals one for workers of race "*x*" and zero otherwise. Female is an indicator variable that equals one for female workers and zero otherwise. Ln(Tenure) is the natural log of the number of quarters that a worker has spent in the SEIN. Manager is an indicator variable equal to one for the highest paid employee in the SEIN and zero otherwise. Multi-unit is an indicator variable equal to one if the firm has more than one establishment and zero otherwise. N_Estabs is the number of establishments owned by the firm, divided by 100. Ln(EstabEmp) is the natural log of establishment employment. Ln(FirmEmp) is the natural log of aggregate firm employment. Ln(# of firms in CT

6.7 Table 5: Labor redeployment - all workers

Read:

- The dependent variable is whether the worker changes two-digit SIC industry after displacement.
- The interaction $Chg\_Q \times Diversified$ is negative and significant.
- A one-standard-deviation increase in future industry Q lowers the probability of industry switching for diversified-firm workers by about 4.1 percentage points relative to focused-firm workers, according to the text.
- The effect remains when same-firm moves and their interactions are included.

Interpretation: The result shows that diversified-firm workers are especially likely to leave declining industries. This is consistent with diversified firms and their workers having broader, more mobile human capital.

Economic message: The external labor market does not reallocate workers across industries as sensitively as the internal labor markets of diversified firms.

6.8 Table 6: Difference in expected growth between new and old industry

Read:

- The dependent variable is the difference in expected growth, measured by the change in Q, between the worker's new and old industries.
- Same-firm moves have a positive coefficient around 0.074.
- The diversified indicator itself is positive but statistically weaker.
- The same-firm coefficient remains stable after year and industry fixed effects and lagged Q controls.

Interpretation: The table asks whether internal switchers move to better industries, not merely different industries. The positive same-firm coefficient suggests that internal labor markets route workers toward industries with stronger prospects.

Economic message: Internal reallocation appears directional and efficiency-enhancing, rather than random reassignment.

6.9 Table 7: Wage changes

Read:

- The dependent variable is the change in annual log wage from before to after establishment closure.
- Changing industries is associated with a large wage loss, around 12–14 percentage points.
- Same-firm moves partly offset wage losses.
- The key coefficient, $Different\ industry \times Spanned\ industry$, is positive and significant, around 0.109 to 0.035 depending on fixed effects.

Interpretation: Workers from diversified firms who move to industries spanned by their former firm experience smaller wage losses. In the strongest SIC-pair fixed-effect specification, the same old-new industry pair is compared, so the result is not merely about choosing better destination industries.

Economic message: The wage evidence supports the view that diversified firms help workers accumulate portable, portfolio-specific skills.

6.10 Figure 3: Worker wage changes by job-change category

Read:

- Focused-firm workers who change industries externally lose about 9.5% relative to the baseline category.
- Diversified-firm workers who move to spanned industries externally lose only about 2.7%.

Table 6
Difference in expected growth (new-old industry)

Dependent variable: Difference in change of Q	(1)	(2)	(3)
Same firm	0.074*	0.074*	0.074*
	(0.042)	(0.042)	(0.042)
Diversified	0.024	0.025	0.025
	(0.017)	(0.017)	(0.017)
Lagged Q			−0.010
			(0.020)
Industry fixed effect	Yes	Yes	Yes
Year fixed effect		Yes	Yes
R^2	0.031	0.032	0.033
N	134,955	134,955	134,955

Table 6 presents estimates from an OLS model that examines the difference in growth rates between the new and original industries of displaced workers who make industry changes. The sample contains one observation for each worker (from focused or diversified firms) whose new job in quarter t + 4 is in a different 2-digit SIC from their original job. The dependent variable is the difference in the change in industry-median Tobin's Q between the worker's new and original industries, where the change in Tobin's Q is measured from year t + 1 to year t + 3 and year t is the year of establishment closure. Same firm is an indicator variable equal to one if the worker is retained within the original firm (firmid) and zero otherwise. Diversified is an indicator variable equal to one if the firm operates establishments in at least two distinct 2-digit SIC codes. Lagged Q is industry-median Tobin's Q for the worker's original industry, measured at the end of the year prior to establishment closure. Standard errors are clustered at the firm level and are reported in parentheses. *, **, and *** represent significance at 10%, 5%, and 1% level, respectively.

Table 7
Wage changes

Dependent variable: $\Delta_{t+4,t-2}\ln(\text{wage})$	(1)	(2)	(3)
Ln(wage)	−0.111***	−0.137***	−0.120***
	(0.009)	(0.009)	(0.008)
Ln(Age)	−0.118***	−0.097***	−0.105***
	(0.010)	(0.009)	(0.009)
Race = Black	−0.040***	−0.045***	−0.035***
	(0.008)	(0.006)	(0.007)
Race = Asian	0.003	0.001	−0.004
	(0.013)	(0.011)	(0.014)
Race = Hispanic	−0.025***	−0.029***	−0.028***
	(0.008)	(0.006)	(0.007)
Race = Other minorities	−0.015*	−0.022***	−0.018**
	(0.008)	(0.007)	(0.007)
Female	−0.041***	−0.051***	−0.035***
	(0.005)	(0.005)	(0.005)
Ln(tenure)	−0.021***	−0.018***	−0.020***
	(0.005)	(0.004)	(0.004)
Manager	−0.001	0.045**	0.009
	(0.020)	(0.021)	(0.020)
N_Estabs	−0.003***		−0.003***
	(0.001)		(0.001)
Ln(EstabEmp)	−0.003	0.012	−0.001
	(0.007)	(0.008)	(0.006)
Ln(FirmEmp)	0.009**		0.013***
	(0.004)		(0.004)
Chg (N_Estabs)	−0.003***	−0.003***	−0.002***
	(0.000)	(0.001)	(0.001)
Chg (EstabEmp)	−0.002	−0.006*	0.000
	(0.003)	(0.003)	(0.003)
Chg (FirmEmp)	0.017***	0.020***	0.016***
	(0.002)	(0.002)	(0.003)
Diversified	−0.006		−0.019
	(0.013)		(0.012)
Same_Firm	0.027*	0.006	0.030*
	(0.016)	(0.023)	(0.016)
Different industry	***−0.144****	***−0.127****	
	(0.017)	(0.022)	
Same_Firm × Different industry	0.052*	0.050	0.062
	(0.030)	(0.042)	(0.039)
Different industry × Spanned industry	***0.109****	***0.104****	***0.038****
	(0.015)	(0.016)	(0.015)
State fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes		
Year fixed effects	Yes	Yes	Yes
Establishment fixed effects		Yes	
SIC pair fixed effects			Yes
R^2	0.112	0.200	0.260
N	42,354	42,354	42,354

The sample contains one observation for each worker displaced from a closing establishment of a multi-unit firm in our matched LBD-LEHD data. Coefficient estimates are from OLS regressions. The dependent variable is the change in the annual real wage from quarter (t − 2) to (t + 4). t is the quarter of the establishment closure. Ln(wage) is the natural log of the annual real wage. Ln(age) is the natural log of the worker's age. Race = "x" is an indicator variable that equals one for workers of race "x" and zero otherwise. Female is an indicator variable that equals one for female workers and zero otherwise. Ln(tenure) is the natural log of the number of quarters that a worker has spent in the SEIN. Manager is an indicator variable equal to one for the highest paid employee in the SEIN and zero otherwise. N_Estabs is the number of establishments owned by the firm, divided by 100. Ln(EstabEmp) is the natural log of establishment employment. Ln(FirmEmp) is the natural log of aggregate firm employment. Chg(N_Estabs), Chg(EstabEmp), and Chg(FirmEmp) are the differences between the old and

- Diversified-firm workers who move to unspanned industries externally lose about 14.3%.
- Internal diversified-firm industry changes perform better than external industry changes.

Interpretation: The tree visually summarizes the central worker-level mechanism. The important contrast is not simply diversified versus focused; it is whether the destination industry is inside the former diversified firm's portfolio. Wage losses are smaller when the worker moves into a spanned industry.

Economic message: Diversified firms build industry-portfolio human capital that can be valuable even after the worker leaves the firm.

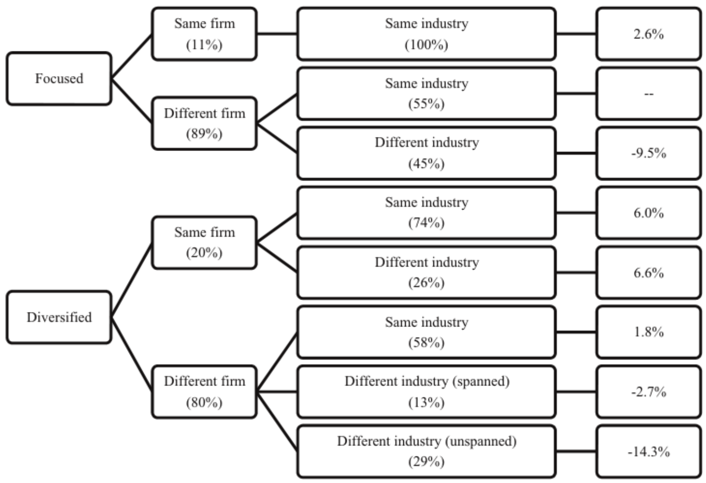

Figure 3
Worker wage changes by job-change category
The sample is workers displaced from closing establishments who are reemployed by the third quarter followi the closure. The baseline job-change category is workers displaced from focused firms who find new jobs in t external market, but remain in their original industries. The figure reports wage losses for the remaining sev job-type categories relative to the baseline. The relative losses are inferred from a linear-regression mirrori the specification in Column 1 of Table 7, but including additional marginal effects for workers from diversifi firms who remain inside their original firm and workers from diversified firms who change industries. All work and firm characteristics are included. The full regression estimates are available in the Online Appendix. At ea node, we report in parentheses the percentage of workers who make each possible type of job change.

6.11 Table 8: Wage changes in high- and low-skill industries

Read:

- The table splits the spanned-industry wage effect by high-skill and low-skill industry measures.
- The high-skill interaction is positive and significant, especially with SIC-pair fixed effects.
- Low-skill interactions are weaker and can disappear after controlling for industry-pair fixed effects.
- The main result is strongest where transferable human capital is most valuable.

Interpretation: This table is a mechanism test. If the wage effect were only about common industry pairs or geography, high-skill intensity should not matter as much. The concentration among high-skill workers supports the skill-development story.

Economic message: The bright side of diversification is especially strong when human capital is complex and transferable.

Table 8
Wage changes in high- and low-skill industries

Dependent variable: $\Delta_{t+4,t-2}$ln(wage)	(1)	(2)	(3)
Ln(wage)	−0.111***	−0.137***	−0.120***
	(0.009)	(0.009)	(0.008)
Ln(age)	−0.118***	−0.097***	−0.105***
	(0.010)	(0.009)	(0.009)
Race = Black	−0.039***	−0.045***	−0.034***
	(0.008)	(0.006)	(0.007)
Race = Asian	0.003	0.001	−0.003
	(0.013)	(0.011)	(0.014)
Race = Hispanic	−0.024***	−0.029***	−0.028***
	(0.008)	(0.006)	(0.007)
Race = Other minorities	−0.015*	−0.022***	−0.018**
	(0.008)	(0.007)	(0.007)
Female	−0.041***	−0.051***	−0.035***
	(0.005)	(0.005)	(0.005)
Ln(Tenure)	−0.020***	−0.018***	−0.020***
	(0.005)	(0.004)	(0.004)
Manager	0.000	0.045**	0.010
	(0.021)	(0.021)	(0.020)
N_Estabs	−0.003***		−0.003***
	(0.001)		(0.001)
Ln(EstabEmp)	−0.003	0.012	−0.001
	(0.007)	(0.008)	(0.006)
Ln(FirmEmp)	0.009**		0.013***
	(0.004)		(0.004)
Chg (N_Estabs)	−0.003***	−0.003***	−0.002***
	(0.000)	(0.001)	(0.000)
Chg (EstabEmp)	−0.002	−0.006*	0.000
	(0.003)	(0.003)	(0.003)

(a) Table 8, part 1

Table 8
Continued

Dependent variable: $\Delta_{t+4,t-2}$ln(wage)	(1)	(2)	(3)
Diversified	−0.006		−0.019
	(0.013)		(0.012)
Same_Firm	0.028*	0.006	0.030*
	(0.016)	(0.023)	(0.016)
Different industry	***−0.146*****	***−0.121*****	
	(0.015)	(0.013)	
Same_Firm × Different industry	0.050*	0.048	0.058
	(0.030)	(0.041)	(0.039)
High_Skill × Different industry	0.004	−0.010	
	(0.026)	(0.034)	
High_Skill × Diff. industry × Spanned industry	***0.117*****	***0.114*****	***0.055*****
	(0.021)	(0.022)	(0.019)
Low_Skill × Diff. industry × Spanned industry	***0.091*****	***0.084*****	***−0.002***
	(0.022)	(0.022)	(0.029)
State fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes		
Year fixed effects	Yes	Yes	Yes
Establishment fixed effects		Yes	
SIC pair fixed effects			Yes
R^2	0.118	0.205	0.260
N	42,354	42,354	42,354

The sample contains one observation for each worker displaced from a closing establishment of a multi-unit firm in our matched LBD-LEHD data. Coefficient estimates are from OLS regressions. The dependent variable is the change in the annual real wage from quarter (t − 2) to (t + 4). t is the quarter of establishment closure. Ln(wage) is the natural log of the annual real wage. Ln(age) is the natural log of the worker's age. Race = "*x*" is an indicator variable that equals one for workers of race "*x*" and zero otherwise. Female is an indicator variable that equals one for female workers and zero otherwise. Ln(tenure) is the natural log of the number of quarters that a worker has spent in the SEIN. Manager is an indicator variable equal to one for the highest paid employee in the SEIN and zero otherwise. N_Estabs is the number of establishments owned by the firm, divided by 100. Ln(EstabEmp) is the natural log of establishment employment. Ln(FirmEmp) is the natural log of aggregate firm employment. Chg(N_Estabs), Chg(EstabEmp), and Chg(FirmEmp) are the differences between the old and new firm in N_Estabs, establishment employment, and firm employment, respectively. Diversified is an indicator variable equal to one for firms that operate in at least two distinct 2-digit SIC codes. Same_Firm is an indicator

(b) Table 8, continued

6.12 Table 9: Worker tenure in the diversified firm

Read:

- The table interacts spanned-industry moves with a low-tenure indicator.
- The baseline spanned-industry effect is positive and significant.
- The triple interaction with low tenure is negative and significant.
- This implies that the benefit of spanned-industry movement is stronger for workers with longer tenure in the diversified firm.

Interpretation: Tenure is crucial because it helps distinguish acquired skills from initial sorting. If workers learn skills inside the diversified firm, longer-tenured workers should have larger spanned-industry wage advantages. Table 9 finds exactly that pattern.

Economic message: Internal labor markets affect not only where workers move, but what human capital they build over time.

6.13 Table 10: Prior experience in the post-displacement new industry

Read:

- The table controls for whether the worker had prior experience in the destination industry before entering the displaced firm.
- Prior experience is not the source of the main spanned-industry effect.
- Long prior experience has a positive direct coefficient in some specifications, but its interaction with spanned-industry movement is not significant.
- The spanned-industry result remains after accounting for prior destination-industry experience.

Table 9
Worker tenure in the diversified firm

Dependent variable: $\Delta_{t+4,t-2}\ln(\text{Wage})$	(1)	(2)	(3)
Ln(wage)	−0.112***	−0.138***	−0.121***
	(0.009)	(0.009)	(0.008)
Ln(age)	−0.117***	−0.098***	−0.104***
	(0.010)	(0.009)	(0.009)
Race = Black	−0.040***	−0.045***	−0.035***
	(0.008)	(0.006)	(0.007)
Race = Asian	0.002	0.001	−0.005
	(0.013)	(0.011)	(0.015)
Race = Hispanic	−0.026***	−0.029***	−0.028***
	(0.008)	(0.006)	(0.007)
Race = Other minorities	−0.015*	−0.022***	−0.018**
	(0.008)	(0.007)	(0.007)
Female	−0.041***	−0.050***	−0.035***
	(0.006)	(0.005)	(0.005)
Ln(Tenure)	−0.026***	−0.012*	−0.024***
	(0.008)	(0.006)	(0.008)
Manager	0.000	0.044**	0.009
	(0.020)	(0.021)	(0.020)
N_Estabs	−0.003***		−0.003***
	(0.001)		(0.001)
Ln(EstabEmp)	−0.003	0.013	0.000
	(0.007)	(0.008)	(0.006)
Ln(FirmEmp)	0.008**		0.011***
	(0.004)		(0.004)
Chg (N_Estabs)	−0.003***	−0.003***	−0.002***
	(0.000)	(0.001)	(0.001)
Chg (EstabEmp)	−0.002	−0.006*	0.000

(a) Table 9, part 1

Table 9
Continued

Dependent variable: $\Delta_{t+4,t-2}\ln(\text{Wage})$	(1)	(2)	(3)
Diversified	−0.006	−0.019	
	(0.013)		(0.012)
Same_Firm	0.043***	0.019	0.045***
	(0.015)	(0.020)	(0.015)
Different industry	***−0.173****	***−0.155****	
	(0.024)	(0.032)	
Same_Firm × Different industry	0.053**	0.065*	0.073**
	(0.026)	(0.038)	(0.036)
Different Industry × Spanned industry	***0.132****	***0.125****	***0.060****
	(0.024)	(0.024)	(0.021)
Low_Tenure	−0.032**	−0.004	−0.022*
	(0.013)	(0.011)	(0.011)
Low_Tenure × Different industry	0.063***	0.059**	0.050***
	(0.021)	(0.025)	(0.013)
Low_Tenure × Different industry × Spanned industry	***−0.058****	***−0.050****	***−0.050****
	(0.025)	(0.025)	(0.022)
State fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes		
Year fixed effects	Yes	Yes	Yes
Establishment fixed effects		Yes	
SIC pair fixed effects			Yes
R^2	0.119	0.190	0.222
N	42,354	42,354	42,354

The sample contains one observation for each worker displaced from a closing establishment of a multi-unit firm in our matched LBD-LEHD data. Coefficient estimates are from OLS regressions. The dependent variable is the change in the annual real wage from quarter (t − 2) to (t + 4). t is the quarter of establishment closure. Ln(wage) is the natural log of the annual real wage. Ln(age) is the natural log of the worker's age. Race = "x" is an indicator variable that equals one for workers of race "x" and zero otherwise. Female is an indicator variable that equals one for female workers and zero otherwise. Ln(tenure) is the natural log of the number of quarters that a worker has spent in the SEIN. Manager is an indicator variable equal to one for the highest paid employee in the SEIN and zero otherwise. N_Estabs is the number of establishments owned by the firm, divided by 100. Ln(EstabEmp) is the natural log of establishment employment. Ln(FirmEmp) is the natural log of aggregate firm employment. Chg(N_Estabs), Chg(EstabEmp), and Chg(FirmEmp) are the differences between the old and

(b) Table 9, continued

Interpretation: Table 10 rules out an important sorting explanation: diversified firms may simply hire workers who already have experience in many of the firm's industries. The evidence suggests instead that skills are developed or strengthened during tenure at the diversified firm.

Economic message: The worker-level mechanism is not just pre-existing match quality; it is also human-capital accumulation inside diversified firms.

7 Interpretive synthesis

- **Diversification creates an internal option.** A diversified firm can move workers across business lines when the marginal product of labor changes across industries.
- **Workers respond to the internal opportunity set.** Because the firm spans multiple industries, workers may invest in broader skill bundles. These skills increase internal redeployability and external opportunities in spanned industries.
- **Productivity and wages can both rise.** Higher wages in diversified firms are not necessarily agency costs. If workers have more valuable skills and stronger outside options, higher wages can coexist with higher productivity.
- **Internal labor markets can change how we interpret internal capital markets.** If diversified firms can adjust labor more flexibly, they may need less capital reallocation to respond to industry shocks. Low capital-Q sensitivity need not automatically imply inefficient socialism.
- **The diversification discount can reflect risk.** If organization capital is embodied in mobile workers, diversified firms may be riskier because workers can leave with part of the firm's human-capital advantage. A discount can coexist with efficient operations.

8 Short summary

- The paper studies the bright side of corporate diversification through internal labor markets.

Table 10
Prior experience in the post-displacement new industry

Dependent Variable: $\Delta_{t+4,t-2}$ln(Wage)	(1)	(2)	(3)	(4)	(5)	(6)
Ln(wage)	−0.112***	−0.138***	−0.121***	−0.112***	−0.138***	−0.122***
	(0.009)	(0.009)	(0.008)	(0.009)	(0.009)	(0.008)
Ln(age)	−0.117***	−0.097***	−0.104***	−0.118***	−0.098***	−0.105***
	(0.010)	(0.009)	(0.009)	(0.010)	(0.009)	(0.009)
Race = Black	−0.040***	−0.045***	−0.035***	−0.040***	−0.045***	−0.034***
	(0.008)	(0.006)	(0.007)	(0.008)	(0.006)	(0.007)
Race = Asian	0.003	0.001	−0.004	0.003	0.001	−0.004
	(0.013)	(0.011)	(0.015)	(0.013)	(0.011)	(0.014)
Race = Hispanic	−0.025***	−0.029***	−0.028***	−0.025***	−0.029***	−0.028***
	(0.008)	(0.006)	(0.007)	(0.008)	(0.006)	(0.007)
Race = Other minorities	−0.016**	−0.023***	−0.019**	−0.016**	−0.023***	−0.019**
	(0.008)	(0.007)	(0.007)	(0.008)	(0.007)	(0.007)
Female	−0.042***	−0.051***	−0.035***	−0.042***	−0.051***	−0.035***
	(0.006)	(0.005)	(0.005)	(0.006)	(0.005)	(0.005)
Ln(Tenure)	−0.021***	−0.019***	−0.020***	−0.020***	−0.019***	−0.020***
	(0.005)	(0.004)	(0.004)	(0.005)	(0.004)	(0.004)
Manager	−0.001	0.044**	0.009	−0.001	0.044**	0.009
	(0.020)	(0.021)	(0.020)	(0.020)	(0.021)	(0.020)
N_Estabs	−0.003***		−0.003***	−0.003***		−0.003***
	(0.001)		(0.001)	(0.001)		(0.001)
Ln(EstabEmp)	−0.002	0.013	0.000	−0.002	0.013	0.000
	(0.007)	(0.008)	(0.006)	(0.007)	(0.008)	(0.006)
Ln(FirmEmp)	0.007**		0.011***	0.007**		0.011***
	(0.004)		(0.004)	(0.004)		(0.004)
Chg (N_Estabs)	−0.003***	−0.003***	−0.002***	−0.003***	−0.003***	−0.002***
	(0.000)	(0.001)	(0.000)	(0.000)	(0.001)	(0.000)
Chg (EstabEmp)	−0.002	−0.006*	0.000	−0.002	−0.006*	0.000
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Chg (FirmEmp)	0.016***	0.019***	0.015***	0.016***	0.019***	0.015***
	(0.002)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)
Diversified	−0.006		−0.019	−0.006		−0.018
	(0.013)		(0.012)	(0.013)		(0.012)
Same_Firm	0.043***	0.017	0.045***	0.042***	0.016	0.045***
	(0.015)	(0.021)	(0.015)	(0.015)	(0.021)	(0.015)
Different industry	***−0.137******	***−0.118******		−0.137***	−0.118***	
	(0.018)	(0.023)		(0.017)	(0.023)	
Same_Firm × Different industry	0.050*	0.070*	0.073**	0.052**	0.072*	0.075**
	(0.026)	(0.038)	(0.035)	(0.027)	(0.038)	(0.035)
Different industry × Spanned industry	***0.103******	***0.102******	***0.035*****	***0.102******	***0.102******	***0.035*****
	(0.016)	(0.015)	(0.017)	(0.016)	(0.015)	(0.017)
Prior experience	−0.030**	−0.040***	−0.014	−0.042***	−0.052***	−0.027*
	(0.015)	(0.013)	(0.014)	(0.015)	(0.013)	(0.014)
Prior exp × Diff. ind. × Span. ind.	***0.000***	***−0.016***	***−0.016***	***0.000***	***−0.014***	***−0.011***
	(0.031)	(0.027)	(0.024)	(0.034)	(0.029)	(0.025)

(*continued*)

4. Discussion

Our analysis documents an important difference between diversified and focused firms. The internal labor markets of diversified firms facilitate

(a) Table 10, part 1

Table 10
Continued

Dependent Variable: $\Delta_{t+4,t-2}$ln(Wage)	(1)	(2)	(3)	(4)	(5)	(6)
Long prior experience				***0.060****	***0.049***	***0.081******
				(0.032)	(0.032)	(0.028)
Long prior exp. × Diff ind. × Span. ind.				−0.026	−0.050	−0.066
				(0.053)	(0.053)	(0.049)
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes			Yes		
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Establishment Fixed Effects		Yes			Yes	
SIC Pair Fixed Effects			Yes			Yes
R^2	0.117	0.190	0.221	0.118	0.190	0.222
N	42,354	42,354	42,354	42,354	42,354	42,354

The sample contains one observation for each worker displaced from a closing establishment of a multi-unit firm in our matched LBD-LEHD data. Coefficient estimates are from OLS regressions. The dependent variable is the change in the annual real wage from quarter (t − 2) to (t + 4). t is the quarter of establishment closure. Ln(wage) is the natural log of the annual real wage. Ln(age) is the natural log of the worker's age. Race = "*x*" is an indicator variable that equals one for workers of race "*x*" and zero otherwise. Female is an indicator variable that equals one for female workers and zero otherwise. Ln(tenure) is the natural log of the number of quarters that a worker has spent in the SEIN. Manager is an indicator variable equal to one for the highest paid employee in the SEIN and zero otherwise. N_Estabs is the number of establishments owned by the firm, divided by 100. Ln(EstabEmp) is the natural log of establishment employment. Ln(FirmEmp) is the natural log of aggregate firm employment. Chg(N_Estabs), Chg(EstabEmp), and Chg(FirmEmp) are the differences between the old and new firm in N_Estabs, establishment employment, and firm employment, respectively. Diversified is an indicator variable equal to one for firms that operate in at least two distinct 2-digit SIC codes. Same_Firm is an indicator variable that equals one if the worker is retained within the firm (firmid) and zero otherwise. Different industry is an indicator variable that equals 1 if the job in quarter t + 4 has a different SIC than the job in quarter t − 2 and zero otherwise. Spanned industry is an indicator variable equal to one if the SIC of the (new) job in quarter t + 4 is an SIC in which the worker's quarter t − 2 firm operates excluding the worker's own industry and zero otherwise. Prior

(b) Table 10, continued

- It uses linked U.S. Census LBD and LEHD data to observe establishments, firms, closures, workers, wages, and worker mobility.
- Diversified firms have higher labor productivity than matched focused firms.
- The productivity advantage is stronger in high-skill industries.
- Establishment closures help isolate involuntary job changes and firm-initiated reallocation.
- Diversified firms retain workers when their remaining industries have better opportunities.
- Diversified firms move workers out of weaker industries and toward industries with stronger Q growth.
- Workers from diversified firms suffer smaller wage losses when moving to industries their old firm also operated.
- The wage advantage is stronger for high-skill workers and workers with longer tenure.
- Prior experience in the destination industry does not explain the effect.
- The results provide a human-capital interpretation of higher wages, productivity differences, internal capital allocation patterns, and diversification discounts.

9 Conclusion

9.1 Core conclusion

Tate and Yang show that diversified firms operate internal labor markets that can create real economic value. Diversification allows firms to redeploy workers across industries and encourages workers to acquire human capital suited to the firm's portfolio of businesses.

9.2 Economic intuition

Workers are not physical assets. They choose skills, respond to career opportunities, and can carry part of the firm's organization capital into the outside market. Diversified firms offer broader internal opportunity sets, which can make workers more productive and more mobile. This can raise firm productivity and reduce wage losses when workers move to spanned industries.

9.3 Incremental contribution in one sentence

The paper's incremental contribution is to show that the benefits of diversification can arise through internal labor markets and worker human-capital accumulation, not only through internal capital markets.

9.4 Takeaway

The paper does not claim that all diversification is efficient. Instead, it adds a missing mechanism to the diversification debate: conglomerates may create value by building and reallocating human capital across business lines. This mechanism helps explain why diversified firms can exhibit higher productivity, higher wages, and possibly valuation discounts at the same time.